



Zoology Legends

Bio202

Mid_Term Papers Spring 2023

Muhammad Zaman 🖱️ 😊

0304-4756496

Join our Legend Group

<https://chat.whatsapp.com/EVU415z0Z4S>

[IgCpc6fBAQX](#)

Team Zoology LEGENDS



Zoology Legends

MCQS (Must Read PPTs)

- ➤ Maleic acid have (**Double Bonds**)
- ➤ It is metabolized to glucose; one gram of GDL yields roughly the same amount of metabolic energy as (**one gram**) of sugar.
- ➤ When glucose is kept in alkaline solution for several hours, it undergoes isomerization to form (**D-fructose and D mannose**).
- ➤ Atomic number of Nitrogen is (**7**)
- ➤ Glycosidic bond of maltose is formed between the OH **of (carbon 1 and carbon 4)** of 2nd glucose monomers. Therefore, it forms an alpha (**1– 4**) glycosidic bond.

Team Zoology LEGENDS



Zoology Legends

- ☞ **(Entropy)** is a measure of the randomness or disorder in a system
- ☞ **(Proline)** residue introduces a destabilizing kink in an α helix
- ☞ Amino acids A. Aliphatic R groups is **(Proline)**
- ☞ The major hemoglobin in adults is **(Hemoglobin A).**
- ☞ This adjustment in the BPG level has **(only a small effect)** on the binding of oxygen in the lungs
- ☞ **(carbon 5 in glucose)** determines whether the sugar belongs to the D or L series
- ☞ Glucose and Galactose are an example of an epimeric pair which differ only with respect to **(C4)**

Team Zoology LEGENDS



Zoology Legends

- **(Eicosanoids)** exert complex control over many bodily systems; in inflammation immunity, and as messengers in the central nervous system.
- ☞ **(Primary systemic amyloidosis)** is caused by deposition of fibrils consisting of misfolded **(immunoglobulin)** light chain
- ☞ Well-defined structures of proteins is **(Tertiary Structure)**
- The amide group of asparagine serves as the site of attachment for an oligosaccharide chain in a glycoprotein... **serine**
- The salt that is water-soluble is...**potassium (K)**.
- The partial pressure of oxygen needed to achieve half saturation of the binding sites is called **p50**.
- Hemoglobin is present in **red blood cells**.

Team Zoology LEGENDS



Zoology Legends

- Immunoglobulins are **proteins** in nature.
- Lauric acid, the saturated fatty acid, has **12** carbon chains.
- Okazaki fragments have **DNA Ligase** Enzyme
- What is the common hemoglobin?
 - Answer: **HbA**
- What is the pressure of oxygen?
 - Answer: **30 mmHg**
- What is a common example of a monosaccharide?
 - Answer: **glucose**
- Which of the following is a polysaccharide?
 - Answer: **Starch**
- What is sickle cell anemia?
 - Answer: a **genetic disorder**
- What is the pKa value of histidine?

Team Zoology LEGENDS



Zoology Legends

- Answer: **6.0**
- What is the pKa value of acetic acid?
 - Answer: **4.76**
- Hemoglobin is a tetra...
 - Answer: **four subunits**
- What is the amount of CO₂ in bicarbonate?
 - Answer: **80 to 85%**
- What are the domains of protein structure?
 - Answer: **Primary, secondary, tertiary, and quaternary**
- Cooperative binding of oxygen refers to the binding of oxygen molecules to **one subunit** of a protein influencing the binding of oxygen to other subunits.
- Are Na and K soluble in water?

Team Zoology LEGENDS



Zoology Legends

- Answer: **Yes**
- What causes protein misfolding?
 - Answer: **Genetic mutations, environmental conditions, or protein imbalances.**
- What is the partial pressure of oxygen in the lungs?
 - Answer: **30 mmHg**
- Glutamate residue is replaced by?
 - Answer: **Valine**
- About ... of the amino acid residues in the myoglobin are found in α helices with bends in between?
 - Answer: **78%**
- What is the pK value of glutamate?
 - Answer: **3.22**

Team Zoology LEGENDS



Zoology Legends

- Is sucrose a reducing sugar?
 - Answer: **No**, sucrose is a non-reducing sugar.
- What is the pKa value of alanine?
 - Answer: **5.7**
- DNA have sugar?
 - Answer: **Deoxyribose**.
- The reappearing process of DNA is called...
 - Answer: **DNA replication**
- A (-PO₃²⁻) from ATP is transferred to an alcohol forming...
 - Answer: **phosphorylation**
- What is the end product of glycolysis?
 - Option: **Pyruvate**
- In the submersion process, organisms grow in which medium?

Team Zoology LEGENDS



Zoology Legends

Option:

- What is the structure of tRNA?
 - Option: **Cloverleaf-shaped structure**
- The synthesis process of glucose is called...
 - Option: **Gluconeogenesis**
- What type of hemoglobin do humans have?
 - Option: **HbA**
- Is histidine polar or non-polar?
 - Option: **polar**
- Mono Amino mono-carboxylic Amino Acid is **Cysteine**
- Hydroxyl group of **serine and threonine** serve the site attachment for Oligosaccharides.
- Which Amino acid in Beta Turn

Team Zoology LEGENDS



Zoology Legends

The designation of a sugar isomer as the D form or as the L form **carbon 5th** in glucose.

- PK2 value of glycine **9.60**
- Which Amino acid acts Both glycolytic / ketogenic... **Lysine / leucine**
- Myoglobin have have **single peptide** Chain
- Enzymes which catalytic / Shuffling is...
Protein disulfide isomerase (PDI).
- Na and K are **H₂O (Water)** Soluble.
- **Increasing** the affinity of the Hb for the further O₂ addition.
- When strenuous exercise lowers the **pO₂** of muscle tissue to about **5 mm Hg**, myoglobin

Team Zoology LEGENDS



Zoology Legends

releases O_2 for mitochondrial synthesis of ATP, permitting continued muscular activity.

- The remainder of the CO_2 is transported as dissolved HCO_3^- -(80 to 85%)
- As in **sickle cell anemia** to deletion of a larger portion of the **polypeptide chain** (as in most cases of Duchenne muscular dystrophy)
- **Maltose** sunflower-shaped
- Thus glucose, fructose and mannose give the same type **(needle-shaped) osazones**.
- Which one is non- polar Aliphatic **proline**
- Nonpolar, **Aliphatic R Groups** includes, Glycine, Alanine, Proline, Valine, Leucine, Isoleucine, Methionine

Team Zoology LEGENDS



Zoology Legends

- Structure of hemoglobin is ? **Tetramal**
- Iron is? **transition metal**
- Sure! Here are the answers to the multiple-choice questions (MCQs) you provided:
- Different compositions of carbohydrate **(oligosacchahride chains)** in different antigens.
- These oligosaccharides are attached to proteins through a **serine or threonine** residue or to ceramide lipid intermediate
- Palmitic acid is a saturated fatty acid and contains **16 carbon atoms**.

Team Zoology LEGENDS



Zoology Legends

- In sickle cell anemia, methionine is replaced by **valine** in the beta-globin chain of hemoglobin.
- The intermediate of glycolysis is called **glyceraldehyde-3-phosphate**.
- Podé puff osazones are given by **reducing sugars** such as glucose and fructose.
- The symbol for tyrosine is **Tyr**.
- E. coli converts lactose to lactate by the enzyme lactate **dehydrogenase**.
- Myosin includes the amino acid called **leucine**.
- Leucine, isoleucine, and tyrosine are amino acids that contain a **heterocyclic R group**.

Team Zoology LEGENDS



Zoology Legends

- The pKa of alanine is approximately **2.34**
- **Histidine** is an amino acid that contains an **aromatic imidazole** group.
- An example of a contractile protein is **actin**.
- Immunoglobulins (antibodies) contain **four protein chains**: two heavy chains and two light chains.
- Stereoisomers that are mirror images of each other are called....**enantiomers**
- Glucose and Galactose are an example of an epimoric pair which differ only with respect to..... **C4**
- ✨ Example of hemopolysacchrides ? Starch

Team Zoology LEGENDS



Zoology Legends

- ✨ Hemoglobin consist of ? Two alpha & two beta chains
- ✨ Classification of protein? All options
- ✨ naturally occurring isomers ? L- isomers
- ✨ maltose consist of which molecules? Two
- ✨ furfural compounds have sugar? Pentose
- ✨ protein examples? All
- ✨ unsaturated fatty acids definition?
- ✨ concentration in lungs and tissues? High
low
- ✨ oligosaccharides?
- ✨ value can be denoted by ? pKa

Team Zoology LEGENDS



Zoology Legends

- ✨ glycosaminoglycans role ?
- 1. ___ is replaced by valine in sickle cell anemia.
- 2. The melting point of fatty acids depend upon
 - a) degree of saturation b) branching of chain.
 - c) both.
- 3. Glutamate is replaced by ___
- 4. Glutathione consists of how many amino acids?
- 5. Enzymes are ___
 - a) proteins b) carbohydrates c) lipids.

Team Zoology LEGENDS



Zoology Legends

Subjective

2, 3 Bisphosphoglycerate (2, 3 BPG)

2, 3 BPG is an alternate product of glycolysis and its concentration increases in RBCs in states of low oxygen delivery to the tissues.

- It is a negatively charged molecule which increases the stability of taut form of hemoglobin.
- 2,3-Bisphosphoglycerate is known to greatly reduce the affinity of hemoglobin for oxygen.
- There is an inverse relationship between the binding of O₂ and the binding of BPG.
- BPG binds at a site distant from the oxygen-binding site
- And therefore regulates the O₂-binding affinity of hemoglobin in relation to the pO₂ in the lungs.

Team Zoology LEGENDS



Zoology Legends

Perks of The Quaternary structure

- The increase in size increases the number of possible interactions between amino acid residues and therefore makes it more difficult for a protein to unfold and refold.
- A multisubunit structure has many advantages besides increased stability.
- It may enable the protein to exhibit cooperativity between subunits in binding ligands (illustrated later with hemoglobin)
- Or it may form binding sites with a high affinity for large molecules (illustrated with antigen binding to the immunoglobulin molecule IgG).
- An additional advantage of a multi-subunit structure is that the different subunits can have different activities and cooperate in a common function.

Team Zoology LEGENDS



Zoology Legends

- For example enzymes that have regulatory subunits or exist as multiprotein complex, such as Pyruvate Dehydrogenase.

- E1 , pyruvate dehydrogenase;
- E2, dihydrolipoyl transacetylase; and
- E3, dihydrolipoyl dehydrogenase

(PDC) is a complex of three enzymes that convert pyruvate into acetyl

CoA by a process called pyruvate decarboxylation

Define disaccharide and provide an example.

Disaccharides are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis

Team Zoology LEGENDS



Zoology Legends

- Maltose - 2 molecules of glucose on hydrolysis
- Lactose – 1 glucose & 1 galactose on hydrolysis
- Sucrose – 1 glucose & 1 fructose on hydrolysis

Describe the basis of protein classification

Different classifications of proteins are based on their:

1. Shape and Size
2. Biological actions/ Functions
3. Solubility and physical properties
4. Quality

Write the characteristics of enantiomers

Team Zoology LEGENDS



Zoology Legends

Stereoisomers that are mirror images of each other are called enantiomers

Enantiomers have nearly identical chemical properties but differ in a characteristic physical property: their interaction with plane-polarized light. Since they are non-superposable mirror images of each other, the two forms represent a class of stereoisomers called enantiomers

Name two diseases caused by the misfolding of proteins.

- Alzheimer's disease
- Type 2 diabetes,
- Huntington's disease and
- Parkinson's disease, arise from a common misfolding mechanism.

What are digestive proteins? Explain with examples.

Team Zoology LEGENDS



Zoology Legends

- These proteins are digestive enzymes which digest our food materials such as carbohydrates, proteins, lipids and include Amylase, Pepsin, Lipases etc.

What are ketogenic amino acids?

- Amino acids whose catabolism yields either acetoacetate or its precursor, (acetyl CoA or acetoacetyl CoA) are termed ketogenic
- Acetoacetate is one of the ketone bodies which also include 3-hydroxybutyrate and acetone
- Leucine and lysine are the only exclusively ketogenic amino acids found in proteins
- Their carbon skeletons are not substrates for gluconeogenesis and, therefore, cannot give rise to the net formation of glucose or glycogen in the liver, or glycogen in the muscles

Team Zoology LEGENDS



Zoology Legends

Provide the characteristics of furfural compounds.

Furfural is an important renewable, non-petroleum based, chemical feedstock.)

The furfural products thus formed can condense with certain organic phenols (a-naphthol) to form compounds having characteristic colors
This forms basis of certain tests used for detection of sugars

Name factors that influence the free energy of protein molecules

- The free energy of a protein molecule is influenced by
- (1) The hydrophobicity,
- (2) Hydrogen bonds,
- (3) Electrostatic interactions, and

Team Zoology LEGENDS



Zoology Legends

- (4) The conformational entropy due to the restricted motion

What do you know about allosteric amino acids?

In biochemistry, allosteric regulation (or allosteric control) is the regulation of an enzyme by binding an effector molecule at a site other than the enzyme's active site.

The site to which the effector binds is termed the allosteric site or regulatory site. Allosteric sites allow effectors to bind to the protein, often resulting in a conformational change involving protein dynamics. Effectors that enhance the protein's activity are referred to as allosteric activators, whereas those that decrease the protein's activity are called allosteric inhibitors.

Team Zoology LEGENDS



Zoology Legends

Explain D and L isomerism with respect to glucose.

The designation of sugar isomer as the D-form or as L-form is determined by its spatial relationship to the parent compound of carbohydrate, the glyceraldehyde.

The orientation of the -Hand-OH groups around the carbon atom adjacent to the terminal primary alcohol or carbon most distant from the carbonyl carbon the penultimate carbon.

Carbon 5 in glucose determine whether the sugar belongs to D or L series when the -OH group in this carbon is on the right the sugar is D-isomer. When it is on the left it is L isomer

List the hydrolysis products of maltose, sucrose, and lactose.

Team Zoology LEGENDS



Zoology Legends

- Maltose - 2 molecules of glucose on hydrolysis
- Lactose – 1 glucose & 1 galactose on hydrolysis
- Sucrose – 1 glucose & 1 fructose on hydrolysis

Explain the cooperative binding of oxygen.

- Cooperative binding of oxygen means that the binding of oxygen to one heme group enables an oxygen binding to the second heme group of the same hemoglobin molecule.
- Conversely, the oxygen release from one heme group facilitates the release of oxygen from other heme groups in the same hemoglobin molecule.

Team Zoology LEGENDS



Zoology Legends

Describe the function of a proton attached to the heme group in tissues and lungs.

Protons bind at various places on the protein, while carbon dioxide binds at the α -amino group. ...

A reduction in the total binding capacity of hemoglobin to oxygen (i.e. shifting the curve down, not just to the right) due to reduced pH is called the root effect.

Function Tissue and Lungs: Hemoglobin transport O₂ from lungs to tissues and CO₂ from tissues to lung.

List the characteristics of histidine

Histidine, which has an aromatic imidazole group

- The side chain of histidine ($pK_a \sim 6.0$) dissociates near physiologic pH, so only a

Team Zoology LEGENDS



Zoology Legends

portion of the histidine side chains carry a positive charge

- Histidine is therefore weakly basic & the free amino acid is mainly uncharged at physiologic pH
- The amino acid side chains might have very different pKas than those of the free amino acids if they are involved in hydrogen or ionic bonds with other amino acid side chains
- Therefore, when histidine is incorporated into a protein, its side chain can be either positively charged or neutral depending on the ionic environment provided by the polypeptide chains of the protein

Discuss the biomedical importance of carbohydrates

- Precursors of nucleic acids: Nucleotides are the building blocks of nucleic acids.

Team Zoology LEGENDS



Zoology Legends

- Without them, DNA or RNA can not be produced.
- Transmission of genetic information;
- This gives them the ability to store and transmit genetic information from;
- one generation to the next which is a fundamental condition for life
- Protein synthesis,
- The ultimate expression of this information, is therefore dependent on nucleotides.
- Energy currency: Nucleotides play an important role as "energy currency" in the cell.
- Nucleoside tri- and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism.
- By doing this, they play a key role in the energy transduction.

Team Zoology LEGENDS



Zoology Legends

- This energy is used in almost every energy requiring process in the body, such as;
- Muscle contraction, Transmission of nerve impulse, Transports of nutrients across cell membrane
 - Motility of spermatozoa And many more energy dependent processes

**Which protein helps in respiration?
Provide an example.**

Heme containing proteins are involved in the function of respiration e.g. Hemoglobin
Myoglobin, cytochromes

How do fireflies produce light?

Team Zoology LEGENDS



Zoology Legends

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase

Provide five examples of proteins that are important in biomedicine.

Describe how proteins work in transport and contraction.

Transport Proteins:

- These proteins transport various substances from one part of the body to the other e.g.
 - Hemoglobin transport O₂ from lungs to tissues and CO₂ from tissues to lungs –
 - Transferrin transports iron

Contractile Proteins:

- These proteins are involved in muscle contraction and relaxation
 - Myosin of thick filaments

Team Zoology LEGENDS



Zoology Legends

– Actin of thin filaments of skeletal muscles

What happens if hemoglobin alliterates in the blood?

- The partial pressure of oxygen needed to achieve half-saturation of the binding sites is called (P50).
- P50 is approximately 1 mm Hg for myoglobin and 26.6mm Hg for hemoglobin
- The partial pressure of oxygen in the blood at which the hemoglobin is 50% saturated, is known as the P50.
- The P50 is a conventional measure of hemoglobin affinity for oxygen.
- Values of P50 are negatively correlated with substrate affinity, with lower values of P50 corresponding to high affinity and vice versa.

Team Zoology LEGENDS



Zoology Legends

- This means that the higher the oxygen affinity (that is the more tightly oxygen binds), the lower the P50

What is starch, and what is its structure?

Starch molecules arrange themselves in the plant in semi-crystalline granules.

Each plant species has a unique starch granular size: rice starch is relatively small (about 2 μm) while potato starches have larger granules (up to 100 μm). Starch becomes soluble in water when heated.

Define entropy

Entropy is a measure of the randomness or disorder in a system.

Entropy of a system always tend to increase to a maximum value.

Define allotropy

Team Zoology LEGENDS



Zoology Legends

The existence of two or more different physical forms of a chemical element

What is HSP60?

- The hsp60 family of chaperones, sometimes called chaperonins, differ in sequence and structure from hsp70 and its homology
- Hsp60 acts later in the folding process, often together with an hsp70 chaperone
- Hsp60 chaperonins form a multi-subunit barrel-shaped structure

Describe the characteristics of furfural compounds.

Furfural is an important renewable, non-petroleum based, chemical feedstock.) The furfural products thus formed can condense with certain organic phenols (a-naphthol) to form compounds having characteristic colors

Team Zoology LEGENDS



Zoology Legends

This forms basis of certain tests used for detection of sugars

How is Duchenne Muscular Dystrophy caused?

- Duchenne Muscular dystrophy results from a large segment of gene deletion in X chromosome, resulting in small truncated dystrophin protein.
- Thus we know that if the primary structure is altered, the function of the protein may also be changed

What is the difference between hemiacetals and hemiketals?

In aqueous solution, aldotetroses and all monosaccharides with five or more carbon atoms in the backbone occur predominantly as cyclic (ring) structures.

Team Zoology LEGENDS



Zoology Legends

The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called hemiacetals or hemiketals.

Which contain an additional asymmetric carbon atom and thus can exist in two stereoisomeric forms

How does misfolding of proteins occur?

- Despite the many processes that assist in protein folding, misfolding does occur.
- Misfolding of proteins may occur spontaneously
- or caused by a mutation in a particular gene, producing an altered protein.

Team Zoology LEGENDS



Zoology Legends

- In addition some apparently normal proteins can, after abnormal proteolytic cleavage, take on a unique conformational state
- That leads to the formation of long fibrillar protein assemblies consisting of β pleated sheets
- Many conditions, including
- Alzheimer's disease
- Type 2 diabetes,
- Huntington's disease and
- Parkinson's disease, arise from a common misfolding mechanism.

What are monosaccharides? Provide two examples.

Monosaccharides (simple sugars) are those which cannot be hydrolyzed further into simpler forms

Team Zoology LEGENDS



Zoology Legends

The backbones of common monosaccharides are unbranched carbon chains in which all the carbon atoms are linked by single bonds

Example: Glucose

What are chaperones, and what is their main function?

- Cells employ auxiliary proteins to overcome these kinetic barriers.
- These auxiliary proteins are called chaperones.
- Chaperone use energy provided by ATP hydrolysis to assist in the folding process
- Chaperones participate in the folding of over half of mammalian proteins.
- Chaperones prevent aggregation, thus providing an opportunity for the formation of appropriate secondary structural elements.

Team Zoology LEGENDS



Zoology Legends

- They prevent immature folding of the nascent polypeptide and help in the final native protein formation.
- The hsp70 family of chaperones binds short sequences of hydrophobic amino acids in newly synthesized polypeptides.
- They also unfold proteins prior to their insertion through the membrane of mitochondria and other organelles.

Define carbohydrates and name four major groups of carbohydrates.

Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis. Many, but not all, carbohydrates have the empirical formula $(CH_2O)_n$; some also contain nitrogen, phosphorus or sulfur.

Team Zoology LEGENDS



Zoology Legends

Classification

- Carbohydrates are classified into four major groups:

1. Monosaccharides
2. Oligosaccharides
3. Disaccharides
4. Polysaccharides

What are anomers and anomeric carbon?

Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers

The carbonyl carbon atom is called the anomeric carbon

What are beta bonds?

Team Zoology LEGENDS



Zoology Legends

Describe the importance of the side chains of alanine, valine, and leucine.

- The side chains of alanine, valine, leucine, and isoleucine tend to cluster together within proteins, stabilizing protein structure by means of hydrophobic interactions.

Explain the concept of a domain in a protein

A protein domain is a conserved part of a given protein sequence and (tertiary) structure that can evolve, function, and exist independently of the rest of the protein chain. Each domain forms a compact three-dimensional structure and often can be independently stable and folded.

Team Zoology LEGENDS



Zoology Legends

The Structure of Proteins

Four levels of protein structure are commonly defined

The Primary Structure

A description of all covalent bonds (mainly peptide bonds and disulfide bonds) linking amino acid residues in a polypeptide chain is its primary structure

The most important element of primary structure is the sequence of amino acid residues

Secondary structure

Secondary structure refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns

Tertiary structure

Tertiary structure describes all aspects of the three-dimensional folding of a polypeptide

Team Zoology LEGENDS



Zoology Legends

Quaternary structure

When a protein has two or more polypeptide subunits, their arrangement in space is referred to as quaternary structure

Provide a note on transcription

- mRNA molecules are formed with the help of DNA template strand (3'-5') during the process called transcription.
- In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated,
- there are untranslated regions at its 5' and 3' ends
- Moreover, there is a 5' cap and
- a poly A tail at 3' end

Team Zoology LEGENDS



Zoology Legends

Explain the quantity structure of proteins.

- Some proteins contain two or more separate polypeptide chains, or subunits, which may be identical or different.
- The arrangement of these protein subunits in three-dimensional complexes constitutes quaternary structure.
- Many proteins have multiple polypeptide subunits (from two to hundreds)
- The association of polypeptide chains can serve a variety of functions
- A multisubunit protein is also referred to as a multimer, with the prefixes —homoll or —heteroll used to describe identical or different subunits, respectively.
- Most multimers have identical subunits

Team Zoology LEGENDS



Zoology Legends

- Or repeating groups of nonidentical subunits, usually in symmetric arrangements.
- The repeating structural unit in such a multimeric protein is protomer
- Assembly into a multisubunit structure increases the stability of a protein.

Define immunoglobulin, transport proteins, and storage proteins.

Immunoglobulins that serve as the first line of defense against bacteria and viruses and other foreign agents are proteins

- The transport proteins carry out the function of transporting specific substances across the membrane or in the body fluids

Team Zoology LEGENDS



Zoology Legends

- Storage proteins bind with specific substances and store them
- Under certain conditions, proteins can be catabolized to provide energy
- Proteins help in the maintenance of water and electrolyte balance in the body by exerting osmotic pressure

What are Glycogenic / Glucogenic amino acids?

- Amino acids whose catabolism yields pyruvate are termed glucogenic or glycogenic
- These intermediates are substrates for gluconeogenesis and, therefore, can give rise to the net formation of glucose or glycogen in the liver and glycogen in the muscle

Team Zoology LEGENDS



Zoology Legends

How does an abnormal sequence of amino acids affect the protein?

An abnormal sequence of amino acids can have various effects on the protein. It can disrupt the protein's three-dimensional structure and alter its folding pattern, leading to misfolding or improper conformation.

This can result in functional impairment, loss of protein stability, and potential aggregation, which may contribute to the development of degenerative diseases.

Moreover, the specific function of the protein can be affected, impacting its ability to interact with other molecules or perform its intended biological role.

Explain the allosteric effect of hemoglobin??

Team Zoology LEGENDS



Zoology Legends

The allosteric effect of hemoglobin refers to the regulation of its oxygen-binding affinity by certain molecules. Hemoglobin exhibits positive cooperativity with oxygen, meaning that the binding of one oxygen molecule enhances the affinity for subsequent molecules.

Various allosteric effectors modulate this process. Oxygen (O_2) positively affects oxygen binding by breaking salt bridges and stabilizing the relaxed form of hemoglobin.

However, molecules like H^+ (protons), CO_2 (carbon dioxide), and 2,3-Bisphosphoglycerate (BPG) negatively affect oxygen binding by stabilizing the tense, deoxygenated form of hemoglobin, reducing its affinity for oxygen.

Explain the steps involved in protein folding??

Protein folding involves several key steps:

Team Zoology LEGENDS



Zoology Legends

1. Translation: Proteins are synthesized as linear chains of amino acids, called polypeptides, through the process of translation.

2. Unfolded State: Initially, the polypeptide exists in an unfolded state or random coil, lacking a stable three-dimensional structure.

3. Folding Initiation: During translation, the polypeptide chain begins to fold into its characteristic three-dimensional structure.

4. Amino Acid Interactions: The amino acids within the polypeptide interact with each other, leading to the formation of specific patterns of secondary structures like α -helices and β -sheets.

5. Tertiary Structure Formation: The secondary structures then fold further and arrange themselves into a specific three-

Team Zoology LEGENDS



Zoology Legends

dimensional arrangement, forming the overall tertiary structure of the protein.

6. Native State: The folded protein achieves its native or functional state, characterized by its unique three-dimensional structure determined by the amino acid sequence.

What is cis configuration in lipids??

In lipids, the cis configuration refers to the arrangement of atoms or groups around a double bond in a fatty acid chain. In the cis configuration, the two hydrogen atoms attached to the carbon atoms in the double bond are on the same side of the carbon chain. This results in a bent or kinked shape in the fatty acid chain.

The presence of cis double bonds introduces flexibility and prevents close packing of lipid molecules, leading to lower melting points

Team Zoology LEGENDS



Zoology Legends

compared to saturated fats and trans fats. Unsaturated fats with cis double bonds are generally considered healthier than trans fats.

Explain the functions of glucose in the body??

Glucose, also known as blood sugar, plays several important functions in the body. It is the primary fuel source for cells, including the brain, red blood cells, and muscles. Glucose is converted into energy through complex biochemical reactions, producing adenosine triphosphate (ATP), which powers cellular activities.

Brain cells rely exclusively on glucose for fuel, while other cells can also use alternative nutrients.

Team Zoology LEGENDS



Zoology Legends

The body obtains glucose from dietary sources, glycogen stores, and gluconeogenesis, the process of manufacturing glucose from other molecules. Insulin helps regulate blood glucose levels by facilitating its entry into cells for energy production

Fructose / Application??

Fructose, also known as fruit sugar or levulose, is a simple ketonic monosaccharide sugar found in many plants, fruits, and vegetables. It is one of the three dietary monosaccharides, along with glucose and galactose, that are directly absorbed into the blood during digestion. Fructose is commonly found in fruits such as mango, litchi, cherry, guava, and vegetables like carrot, radish, and beetroot. It is obtained commercially from sugar cane, maize, and sugar beets.

Team Zoology LEGENDS



Zoology Legends

Applications of fructose include its use in the production of high-fructose syrups (HFS) which are widely used in various food and beverage applications.

HFS, containing fructose, dextrose, and minor amounts of oligosaccharides, provides several physical and functional attributes to food and beverages. These attributes include sweetness, flavor enhancement, humectancy, color and flavor development, freezing-point depression, and osmotic stability.

HFS is extensively used in carbonated beverages, baked goods, canned fruits, jams and jellies, and dairy products. Crystalline fructose and crystalline fructose syrup are also used in a wide range of food and beverage applications.

Team Zoology LEGENDS



Zoology Legends

Describe the functions of auxiliary proteins??

Auxiliary proteins have important functions in different biological processes. In the context of Rubisco, an enzyme involved in photosynthesis, auxiliary proteins are known to play a role in Rubisco biogenesis and function.

In the case of AMPA receptors, which are involved in synaptic transmission in the brain, auxiliary proteins modulate the functional properties of AMPA receptors and/or mediate surface trafficking [3].

How does the alpha helix stabilize tautomers??

Describe the biological importance of carbohydrates.??

L Rhamnose and L fucose short tha

Team Zoology LEGENDS



Zoology Legends

1 short tha lipids waly topic sy 2,3
phospho something.. above 109 module

Configuration for planer peptide bond

Hydrolysis product of maltose,sucrose
and lactose

Cooperative binding of oxygen

Function of proton attached to heme
group in tissue and lungs

Characteristics of histidine

Biomedical importance of carbohydrates

Steps involved in protein folding

Properties of D fructose

Write note on Beta bends? Marks5

Glutamine characters?

Tetrative cycle of glycine?

Team Zoology LEGENDS



Zoology Legends

Binding of hemoglobin at sea level and at high altitudes?

Equilibrium constant?

sorbitol ki importance?

Methionine, alanine, arginine inky first letter recognition?

Cooperative binding of protein?

Ring structure tha tyrosine tryptophan or phynalaline?

proton binding?

dessication of amino acid of alanine?

Team Zoology LEGENDS



Zoology Legends

what are aromatic amino acids give three examples?

Beta bends part of secondary structure?

E. Negatively Charged (Acidic) R Groups

Negatively charged (acidic) R groups: Amino acids with negatively charged (acidic) R groups are proton donors and contain a negatively charged carboxylate group ($-\text{COO}^-$) at neutral pH. Examples of amino acids with negatively charged R groups are aspartate and glutamate, which also have a second carboxyl group.

Team Zoology LEGENDS



Zoology Legends

Physiological pH: At physiological pH (pH 7.0), the side chains of these amino acids are fully ionized, resulting in a net negative charge.

Essential amino acids: Humans have no dietary requirement for protein as a whole, but the protein in food provides essential amino acids. Among the 20 amino acids needed for protein synthesis, nine are considered essential because the body cannot synthesize them at an adequate rate. These essential amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

Team Zoology LEGENDS



Zoology Legends



Muhammad Zaman

03044756496

Team Zoology LEGENDS



Zoology Legends

Remember me in you Precious Prayer

Jazak Allah



ZOOLOGY
LEGENDS

Team Zoology LEGENDS